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Characteristics of dust aerosols and identification of dust sources in Xinjiang, China

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HIGHLIGHTS

· Daily dust aerosol optical depth (DOD) data was obtained from 2003 to 2019
· Anthropogenic and natural dust sources of Xinjiang, China, were identified
· Dust aerosols emissions in dust sources were quantitatively estimated
· The influences of precipitation, topography, and human on dust aerosols were analyzed

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ABSTRACT

Changes in atmospheric particulate matter content directly affect the Earth-atmosphere system balance and climate change and indirectly affect the environment and ecology. Therefore, this matter is receiving increasing attention. A typical arid area with a fragile ecological environment and frequent dust activities, Xinjiang, China, was investigated using Moderate Resolution Imaging Spectroradiometer (MODIS) daily aerosol optical thickness and Ångström wavelength exponent data to obtain the temporal and spatial distribution characteristics of long-term dust aerosols over 20 years. The frequency of occurrence was calculated to accurately identify anthropogenic and natural sources, and data, such as wind speed, were superimposed to quantitatively estimate dust aerosol emissions. The factors that influenced dust aerosol emissions are discussed. The results show that the distribution of dust aerosols in South and North Xinjiang is obviously different, and the average dust aerosol optical depth (DOD) in spring is higher than that in other seasons. The sources of natural dust have little variation and are primarily located in Lop Nur, on the edges of the Taklamakan and Gurbantunggut deserts, in the central Tuha Basin, and in dried-up lakes. The sources of anthropogenic dust are mainly distributed in the farmland areas of the oasis plain and the impact fan plain. Natural dust aerosols are dominant in Xinjiang, with high natural dust emissions in spring. Changes in anthropogenic dust aerosols are caused by production activities, and the annual average anthropogenic dust emissions account for 8% of the total dust emissions. Precipitation can promote a reduction in natural dust emissions. Topography has an obvious effect on the formation and transportation of dust aerosols, which manifests as the frequent occurrence of dust in the basin. The population density data and Defense Meteorological Satellite Program/Operational Linescan System (DMSP/OLS) nighttime light data reveal, that the intensity of human activities has a great impact on anthropogenic dust aerosols. This study provides support for research on how aerosols affect climate and environmental changes in Xinjiang.

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1. Introduction

Aerosols are the general name of solid and liquid colloids suspended in the atmosphere, and aerosols affect the radiation budget of the Earth-atmosphere system by absorbing and scattering solar and terrestrial radiation, which affect the local and even global climate (Fernández et al., 2017; Perrone et al., 2015; Garrett, 2003). Dust aerosols account for approximately 25% of the total aerosols(Kinne et al., 2006; Sreekanth, 2014a), have excellent absorptivity, directly affect the radiation budget and heat balance of the Earth-atmosphere system, and accelerate the melting speed of surface snow. Dust aerosols also carry elements, such as N and Fe, to the sea via long-distance transmission, causing eutrophication, which in turn affects the global carbon cycle (Bishop et al., 2002; Guo et al., 2017). Global dust sources are widely distributed and primarily located in arid and semiarid areas, such as the Sahara Desert, Central Asia, the Middle East, North America, and Australia, which provide hundreds of millions of tons of dust to the atmosphere. There is loose surface soil and sparse vegetation in arid areas with, dry climates, which provide a rich material basis for the formation of dust (Liu et al., 2020; Chen et al., 2021). The large amount of dust emissions has significant impacts on air quality, human health, the ecological environment and the global environment (Naeger, 2018; Parajuli and Yang, 2017). Therefore, it is necessary to accurately identify the sources of dust aerosols and quantify the amount of dust aerosol emissions to provide accurate data support for evaluating the effects of aerosols on climate change. Meng et al. (2020) simulated transport structures and budgets of dust aerosols using the Weather Research and Forecasting with Chemistry (WRF-Chem) model in the Tarim Basin, Northwest China and found that the sources of dust emissions were centered over the northeastern Tarim Basin, with a high dust emission flux reaching 24 μg m^{-2} s^{-1}. Pan et al. (2019) studied the impact of dust aerosols on the microphysical properties of cirrus over the Taklimakan Desert in Northwest China. Capelle et al. (2018) retrieved dust aerosol optical depth based on infrared observations and compared the result with the Aerosol Robotic Network (AERONET). Zhang et al. (2017b) collected 40 samples based on the spatial distribution of dust geochemical elements in the Hexi Corridor Desert to analyze the source of dust in the Hexi Corridor, and found that surface characteristics affect dust emissions. Ginoux et al. (2012) ascribed dust sources to natural and anthropogenic origins based on Moderate Resolution Imaging Spectroradiometer (MODIS) Deep Blue (DB) estimates of dust optical depth in conjunction with other data sets, including land use, and calculated that anthropogenic sources accounted for 25% of emissions. Liu and Liu (2015) used an improved algorithm (dynamic reference brightness temperature differences, DRBTD) to study the temporal and spatial distribution of dust storms in northern China and Mongolia from 2000 to 2010 and found that the probability of dust events occurring in spring exceeded 63.8%. Wang et al. (2004) collected dust samples released by different land types in the Heihe River Basin and found that dry lake beds and degraded grasslands had the highest amount of dust released, 29.37 g day^{-1} and 23.54 g day^{-1}, respectively, which were 10 times other land types.

Xinjiang is a typical arid and semiarid area with an arid climate, low precipitation, sparse vegetation, and strong wind, and the surface of the earth is extremely susceptible to wind erosion, which results in high dust emissions. The ecological environment is very fragile in Xinjiang, and frequent dust activities pose a very large threat to the ecological environment in arid areas (Shen et al., 2020). Previous studies of dust emissions have been based on statistical data from weather stations and natural dust, and most of the research has been limited to the Taklimakan Desert. However, long time series and monitoring of the entire Xinjiang region have seldom been carried out. The effects of human activities and natural changes on dust aerosols are not clear. Xinjiang is the core area of the Belt and Road, and it is an area focused on development and construction. The accurate identification and monitoring of anthropogenic and natural dust sources and emissions as soon as possible helps to maintain the ecological security of Xinjiang and supports research on how aerosols affect Xinjiang's climate and environmental changes. The present paper used long-term MODIS data and reanalysis data to obtain the spatial-temporal distribution characteristics of dust aerosol optical depth over the past 20 years in Xinjiang, and the frequency of occurrence of dust was calculated, to further accurately identify anthropogenic and natural dust emission sources, quantitatively assess dust emissions from the main dust sources in the region, and clarify the impact of various factors on dust aerosols.

2Methods.

2.1. Study region

Xinjiang, China, is situated in the center of the Eurasian continent where the climate is dry and windy and affected by the atmospheric circulation systems of the westerly belt. Because the Tienshan Mountains extend from east to west across the vast study area, the climate difference between North Xinjiang and South Xinjiang is obvious (Fig. 1). Approximately 53% of areas are at elevations of 500–1500 m, and local depressions in Xinjiang are below 500 m (Turpan Aydingkol Lake is at an elevation of −154 m, the second lowest point in the world). Sandy deserts and gravel gobi deserts are widely distributed, with an abundance of dust materials and rich wind power resources (Wang et al., 2020; Yao et al., 2020). There are strong wind days in the desert area during spring, and strong wind erosion has become the source of dust and salt dust (Chen et al., 2020; Zhang et al., 2017a). The ecosystem of Xinjiang is fragile and extremely sensitive to climate change. Frequent dust activities have aggravated the fragility of the ecological environment and seriously threaten the ecological security of arid areas.

2.2. Data sets

2.2.1. MODIS

MODIS on the Aqua satellite is widely used in atmospheric remote sensing monitoring. The atmospheric aerosol parameters retrieved by the DB algorithm (Hsu et al., 2013) that are suitable for arid regions are used in this paper. The aerosol optical depth (AOD) data and Ångström wavelength index (α) of the MODIS daily products, 17 level 2, at 550 nm with a spatial resolution of 0.1 ° × 0.1 ° were selected. The Interactive Data Language (IDL) programming tool was used to project and mosaic the 17 years of daily data, and ArcGIS was used to complete the

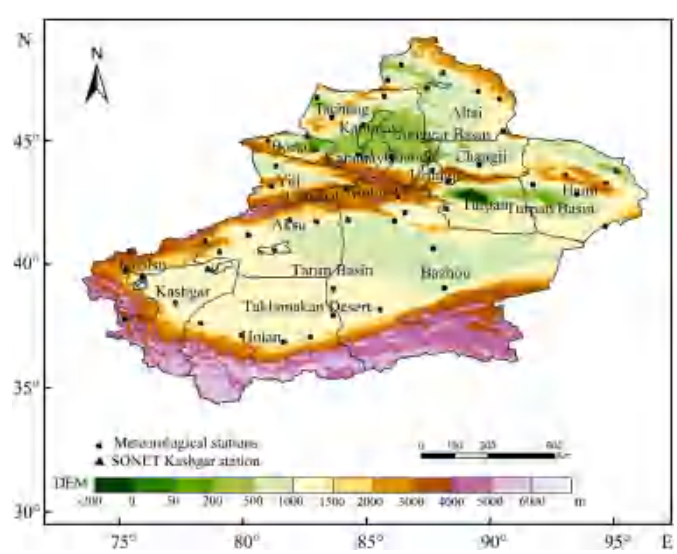

Fig. 1. Elevation of the study area and distribution of stations.

preprocessing processes, such as clipping. The products are available on the NASA website (https://search.earthdata.nasa.gov/search).

2.2.2. Ground-based observation data

The Sun-sky-radiometer Observation Network (SONET) is a ground-based aerosol remote sensing network that was established by the Chinese Academy of Sciences and many Chinese universities and research institutes (Li et al., 2018). SONET obtains high-precision aerosol characteristic parameters, which were used for the verification of satellite remote sensing products and the establishment of aerosol characteristic models. The observation equipment used was primarily CE318-DP (Wei et al., 2020). The L1.5 data, which removed the influence of clouds from the Kashgar site, were selected for comparisons with the satellite data and are available on the SONET website (www.sonet.ac.cn).

MICROTOPS II is a handheld solar photometer that obtains AOD data at 8 different wave bands. The observation point of the solar photometer was set up in Urumqi, the daily observation period was 12:30–16:00, and the sampling interval was 15 min.

The measured aerosol parameters in this study were obtained from ground-based observation stations in Urumqi (northern Xinjiang) and Kashgar (southern Xinjiang) (site distribution is shown in Fig. 1, and site information is shown in Table 1).

2.2.3. Land use and land cover

The International Geosphere-Biosphere Project (IGBP) classification scheme of MCD12Q1 data at a spatial resolution of 500 m was selected to assist in obtaining the underlying surface type of the dust sources. Eleven images covered the research area and were mosaiced and reprojected using HEG tools.

2.2.4. Digital elevation model

The digital elevation model (DEM) data set from the Shuttle Radar Topography Mission (SRTM) was used to analyze the impact of elevation on dust aerosols and may be downloaded from http://gdex.cr.usgs.gov/gdex/, then preprocessed with ArcGIS.

2.2.5. DMSP/OLS nighttime light data

The DMSP/OLS sensor works at night and detects low-intensity light emitted by city lights, small-scale residential areas and traffic. Therefore, DMSP/OLS nighttime light images may be used as a representation of human activity and have become a good data source for human activity monitoring research (SMALL et al., 2005). DMSP/OLS data at a spatial resolution of 1 km from 1992 to 2013 are available. Data from 2003 to 2013 were selected as indicators of human activities to analyze the relationship between human activities and dust aerosols in this study. The light index ranges from 0 to 63, and the value of 0 represents the background value.

2.2.6. ERA-Interim

ERA-Interim contains atmospheric parameters (e.g., temperature at different pressure levels, relative humidity, and U and V components of wind speed.) and surface parameters (e.g., soil moisture content and sea surface temperature) from 1979 to the present. Research results show that the ERA-Interim meteorological data have good applicability in China (Wang et al., 2015). The horizontal wind speed (U and V components), air pressure, and temperature data with a temporal resolution of one month and spatial resolution of $0.125^\circ \times 0.125^\circ$ were selected to calculate the average annual dust emissions.

Table 1
Information on observation sites.

Observation point	Location	Time
Kashgar	39.5°N, 75.9°E	2013-06-30 - 2019-12-18
Urumqi	43.7°N, 87.6°E	2019-03-25 - 2019-07-16

2.2.7. Population density data

The World Population Grid Dataset is a population distribution grid data set (quantity and density) that combines social, economic and remote sensing data, and it is widely used in research to solve problems related to humans and the global environment (Abson et al., 2012; Ermert et al., 2013). GPWv4 population density data with a spatial resolution of 2.5 min (equivalent to 5 km at the equator) were used to analyze the impact of human activities on sand and dust aerosol emissions.

2.3. Methodology

2.3.1. Dust aerosol extraction

Based on ground station and satellite remote sensing data, the most common methods used to distinguish dust aerosols are the atmospheric AOD and Ångström wavelength index (α) (Bibi et al., 2016; Pathak et al., 2012; Kaskaoutis et al., 2009). The AOD describes the aerosol-induced reduction in light, and the Ångström wavelength index is the proportion of large and small particles in the aerosol composition. Sreekanth (2014b) and Kumar et al. (2015) defined particles with a small Ångström wavelength index ($\alpha < 0.7$) as dust aerosols and studied the characteristics of dust aerosols in different regions of the world. Using DOD extraction methods from Sreekanth and Kumar et al., the threshold value of the Ångström wavelength exponent was used as a condition for screening, and the AODs that met the condition were extracted as DODs. This article uses the same method to extract the dust aerosol optical depth (DOD).

2.3.2. Calculation of dust emissions

Due to the sparse ground stations in Xinjiang and the lack of input parameters, it is difficult to simulate the emissions of sand and dust aerosols. However, satellite aerosol data may be used to solve this problem. Based on the ERA-Interim data reanalysis and the frequency of dust occurrence data, the method of Ginoux et al. (2001) was used to estimate the dust emissions, i.e., FP. The calculation formula is shown in Equation (1):

$$F_{\mathrm{p}} = \mathrm{CSu}_{10\mathrm{m}}^{2}(u_{10m} - u_{\tau}) \tag{1}$$

where C is a dimensional factor with a value of 1 $\mu g^2\ m^{-5}$, S is the proportion of dust sources (frequency of dust occurrence), u_{10m} is the wind speed at 10 m, and u_t is the wind erosion threshold.

3. Results

3.1. Data accuracy verification

The AOD at 550 nm from the SONET and the MICROTOPS II handheld solar photometer site was deduced using to the Ångström wavelength index formula (Equation (2)) (Ångström, 1964), which matched the AOD at 550 nm of MODIS.

$$\alpha = -\frac{\log(\tau_1/\tau_2)}{\log(\lambda_1/\lambda_2)} \tag{2}$$

where λ_1 and λ_2 represent two different wavelengths and τ_1 and τ_2 are the AODs at the corresponding wavelengths. The DOD data were extracted from the ground-based AOD data after wavelength matching using the method mentioned above. The ground-based data are continuous observation values at a certain point, and the satellite data are instantaneous observation values on the surface. Therefore, the data needed to be matched in time and space. Using the time of Aqua satellite passes as the center, the average of the ground-based data between 1:00 p.m. and 2:00 p.m. local time was calculated to match the observation values at the time of satellite passes. Using the location of the observation station as the center, the average MODIS data within a range of 30 km were calculated to match the observation values at the location.

Linear fitting analysis was performed on the matched data by establishing a linear regression equation and obtaining the statistical indexes of the correlation coefficient R, root mean squared error (RMSE), relative mean bias (RMB), mean absolute error (MAE) and expectant error (EE) to evaluate the applicability of MODIS aerosol data and the accuracy of DOD extraction methods(Levy et al., 2013).

A total of 860 pairs of data (49 pairs at Urumqi station and 811 pairs at Kashgar station) were matched using the spatiotemporal matching method, and the comparison results of the station and satellite data are shown in Fig. 2. There was good agreement between the AOD data from the Urumqi site and MODIS. The correlation coefficient and the slope of the fitted line were 0.790 and 0.909, respectively. The RMSE (0.155) and MAE (0.109) were relatively small, and 61.22% of the samples were within the expected error interval. There was a slight overestimation (RMB = 1.390).

This situation may be related to the surface heterogeneity of the city, which caused a reduction in the accuracy of surface albedo estimation (Wang et al., 2017), or it may be an error caused by the limited observation data. At the Kashgar SONET site, the correlation coefficient R between the MODIS and SONET AOD data was 0.781, which showed good consistency, but half of the samples were lower than the EE (272 pairs of data are within the EE range). The satellite AOD product significantly underestimated the AOD (the RMB value was 0.903).

The MODIS DB algorithm overestimates surface reflectivity and aerosol single scattering albedo in areas where coarse mode aerosols are dominant, which leads to error (Bilal et al., 2016). In summary, MODIS AOD products achieved good verification accuracy in Xinjiang. Although there were some differences between some samples, the overall trend was consistent. Several studies confirmed the accuracy of MODIS aerosol products in the Xinjiang region (Li et al. 2012, 2017; Xia et al., 2021). We verified MODIS and ground-based α data during the entire period and all samples. The results are shown in Fig. 3. The slope of the fitted lines of the MODIS and ground-based α data was 0.72, and the correlation coefficient R was 0.64, which was overall lower than the coefficient for the AOD products. Most of the scatter lies above the 1:1 line, which suggests that the α of MODIS is overestimated. The α product of MODIS is applicable in the Xinjiang region despite some deviations from the station-based α. Therefore, the MODIS data are reliable for studying changes in aerosol optical characteristics in Xinjiang.

Linear fitting analysis was performed on the daily DOD data from MODIS and the ground stations (Fig. 4), and the results showed a significant correlation between the MODIS DOD and the DOD of the ground station (R = 0.905), but the stability of the fitted line was poor (RMSE = 0.655, MAE = 0.472), which may be due to the limited number of verification data points. These results indicate that the conditions of α < 0.7 and AOD>0.2 imposed on MODIS AOD can identify dust events and eliminate the influence of fine mode pollutants.

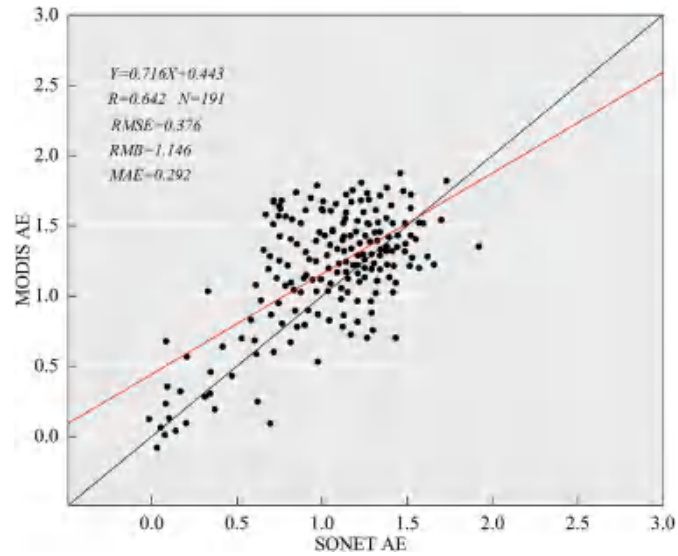

Fig. 3. Comparison of α data from MODIS and ground-based.

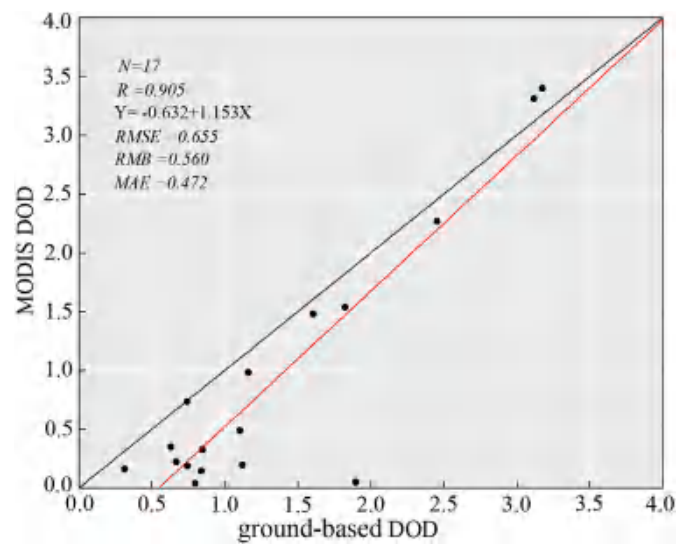

Fig. 4. Comparison between MODIS DOD and SONET DOD

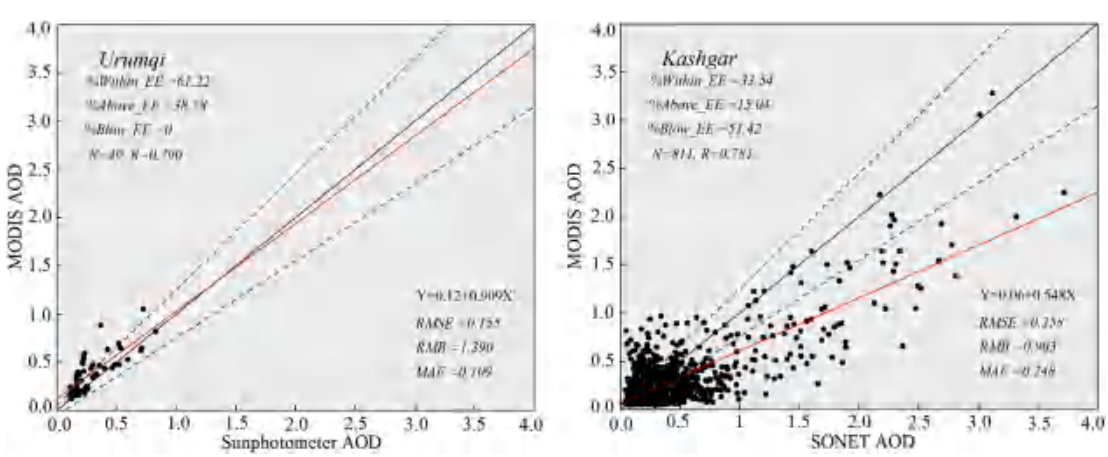

Fig. 2. Comparison of AOD from MODIS, SONET and the sun photometer site.

3.2. Spatial and temporal distribution characteristics of DOD

3.2.1. Spatial distribution characteristics

The dust aerosol optical depth was obtained using the dust aerosol extraction method, the daily aerosol optical depth data and the Ångström wavelength index, and the annual average DOD in Xinjiang from 2003 to 2019 was calculated (Fig. 5). There were significant differences in the distribution of dust aerosols between southern and northern Xinjiang. North of the Tianshan Mountains, the areas with high values of DOD primarily appeared in the center of Hami and Turpan, and the annual average AOD value in regions with DOD values greater than 0.1 was higher than 0.38. South of the Tianshan Mountains, the distribution of DOD was primarily in the Tarim Basin and its surrounding areas. There were high DOD values in Hotan County and southern Aksu located in the oasis on the edge of the Taklimakan Desert. The maximum DOD values in Ruoqiang County reached 0.21. The Tarim Basin is surrounded by mountains on three sides, with low vegetation coverage, sparse precipitation, and frequent dust weather, which caused the high DOD values.

The seasonal distribution of DOD from 2003 to 2019 in Xinjiang is shown in Fig. 6. Overall, the DOD values were relatively high in Xinjiang in spring, especially in the central and southwestern deserts of South Xinjiang, where the average DOD exceeded 0.25. Larger DOD values in North Xinjiang appeared in the central Turpan Basin and southwestern Zhunggar Basin, and the DOD value in the Ebinur Lake Basin was 0.23. The special geographical location and natural environment of the Ebinur Lake Basin make it a typical salt dust aerosol source area. Spring is the warming period in Xinjiang and the peak period of frequent transmission of dust aerosols. Therefore, coal-burning pollutants and dust affected the air quality in spring. Increased precipitation in summer effectively reduced the content of atmospheric dust aerosols, and the range and distribution of DOD in the entire region were significantly lower than spring. The DOD values in most of Xinjiang were the lowest in autumn. The DOD decreased on a large scale and primarily existed in the Aksu area and Ruoqiang County. Humidity decreases and cold air invades frequently during this season, which are conducive to the diffusion of atmospheric pollutants. The lack of regional aerosol data in winter was due to partial snow cover of the ground surface, which did not satisfy the inversion algorithm and resulted in a low seasonal average DOD. However, the range and distribution were larger than these factors in autumn. Compared to autumn, the distribution range of DOD in South Xinjiang also increased significantly in winter. The valley topography caused a concentration of pollutants, which resulted in a higher DOD around the South Xinjiang desert.

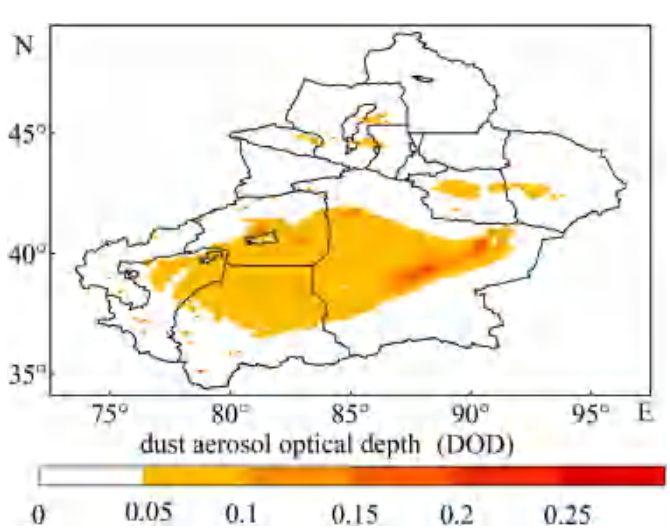

Fig. 5. Spatial distribution of annual mean DOD.

3.2.2. Temporal distribution characteristics

The monthly average changes in AOD and DOD in different regions of Xinjiang from 2003 to 2019 are shown in Fig. 7. The changes in southern and northern Xinjiang during the year were similar and exhibited a unimodal distribution. The annual average values of AOD and DOD in North Xinjiang were relatively low, and the variations were relatively stable. The maximum AOD and DOD values appeared in March and were 0.26 ± 0.14 and 0.03 ± 0.05, respectively. From June to September, AOD and DOD in North Xinjiang showed a downward trend due to the reduction in dust weather and the increase in rainfall, and the lowest AOD and DOD values occurred in October. The AOD and DOD values in South Xinjiang fluctuated significantly during the year, and the monthly average values were greater than North Xinjiang. The AOD and DOD values in South Xinjiang increased rapidly in February and remained at a relatively high level in the spring. The maximum AOD and DOD values appeared in April and were 0.68 ± 0.27 and 0.10 ± 0.08, respectively, which indicates that spring dust aerosols have a significant impact on the air quality in southern Xinjiang.

Fig. 8 shows the annual average AOD and DOD in different regions of Xinjiang from 2003 to 2019, and the regional differences are obvious. The annual average AOD value in Xinjiang was 0.25, which was basically identical to the average AOD value in China (0.26) in the existing research. The average range of DOD in the study area was between 0.01 and 0.05. The fluctuation range of AOD and DOD in South Xinjiang was obvious, and the average values were higher than the other regions and all of Xinjiang. The average annual AOD and DOD changes in North Xinjiang were relatively stable. The annual average AOD was between 0.13 and 0.17, and the DOD changed in the range of 0.01–0.02.

3.3. Identification of the dust source

3.3.1. Frequency distribution

A dust source can be identified by calculating the frequency of occurrence (FOO) of dust aerosols (Prospero, 2002). The basic data for this method are instantaneous remote sensing data. At the observation time, the observed dust is newly discharged dust (dust settles rapidly due to gravity), and the high incidence areas of newly discharged dust are often located in the source of the dust. The seasonal distributions of dust FOO (DOD>0.2) are shown in Fig. 9. The FOO of dust in the Tarim Basin, Zhungeer Basin and Turpan Basin was greater than 5% in spring. The highest FOO of dust occurred in the area around Aiding Lake in Turpan and Shanshan County, with a frequency of more than 25%. The FOO of dust in parts of Changji, Hami and Jinghe Counties was greater than 10%. The high-incidence dust areas in South Xinjiang were in Lop Nur and the Kumtag Desert in northern Ruoqiang County, with the highest FOO reaching 23%. The FOO in part of Yuli County and Aksusaya County was between 15% and 20%. The range of dust activities in the whole area was significantly reduced in summer. The FOO in Shanshan County, Turpan, and Ruoqiang County in Bazhou dropped to approximately 17%, but the FOO in the Aibi Lake Basin was similar to that in spring at approximately 12%. Sandstorms in Xinjiang decreased in autumn and winter, and the FOO was greater than 15% only in the Lop Nur area in Ruoqiang County, which is a high incidence area of dust throughout the year. The FOO in the Zhungeer Basin during winter was extremely small, which may be related to the abundant snowfall during winter.

The average annual dust days from 2003 to 2019 (Fig. 10) were obtained from the dust aerosol data, and the spatial difference in the frequency of dust was analyzed. The number of dust days in most areas of Xinjiang was less than 10 days. Dust activities primarily occurred in the Turpan Basin area and the Taklimakan Desert in South Xinjiang and its surrounding areas.

3.3.2. Natural and anthropogenic dust sources

In arid and semiarid areas where vegetation cover is sparse and sandstorms frequently occur, humans directly or indirectly affect the

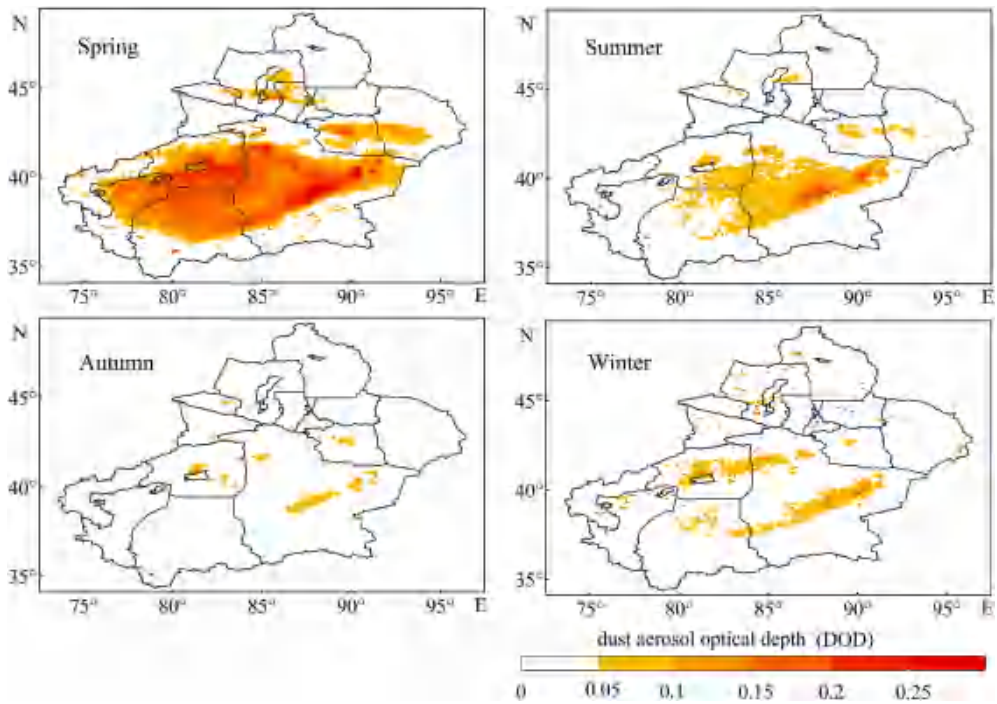

Fig. 6. Spatial distribution of the average DOD in different seasons.

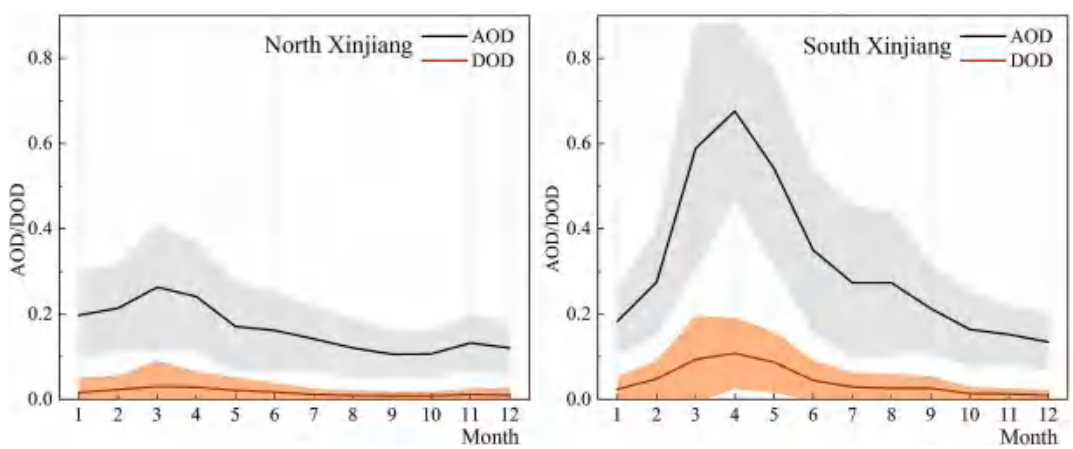

Fig. 7. Intra-annual variations in AOD and DOD in the Xinjiang region.

formation of dust by changing land use and land cover. Therefore, studying the relationship between the types of land use/cover and dust activities can explain the formation of dust to a certain extent. Using MCD12Q1 land type data and surface classification methods from Huang et al. (2015) and related studies, the main human activity areas (e.g., cities, farmland, and grassland) are defined as anthropogenic surfaces, and the remaining types are defined as natural surfaces. Therefore, the surface classification method was used to identify anthropogenic and natural dust sources in the Xinjiang region. The frequency distribution of dust in anthropogenic and natural surface areas during different time periods was analyzed.

The distribution map of anthropogenic and natural dust FOO from 2003 to 2019 (Fig. 11) shows that areas with natural dust FOO ranging from 10% to 15% (the average annual number of dust days is between 37 and 55 days) were primarily distributed in the Lop Nur area, the northwest and southeast edges of the Taklimakan Desert, the Gurbantunggut Desert and the central area of the Tuha Basin. The range of anthropogenic dust sources was significantly smaller than that of natural dust sources. Anthropogenic dust sources were primarily distributed in the oasis plains around the Turpan and Tarim Basins and the piedmont alluvial fan plains on the northern slope of the Tianshan Mountains; The FOO in cultivated land areas in parts of Turpan was greater than 10%.

According to the distribution of the frequency of occurrence of dust aerosols on different surfaces, natural dust sources were primarily distributed in deserts, gobi areas, salt lakes, dry lakes, and dry rivers. The surface cover changes in these areas are weak, and the dust emissions are relatively stable. The ecology of densely populated oases on the edge of deserts is originally fragile, and strong winds and frequent changes in land cover types caused by human activities there make the area a new source of dust. However, the sand source formed by human disturbance can be restored in a short period of time. Greater vegetation coverage induces the stronger hindering effects of vegetation on sand and strengthens resistance of surface materials to wind erosion. Therefore, the occurrence of dust frequency decreases quickly as the surface vegetation increases.

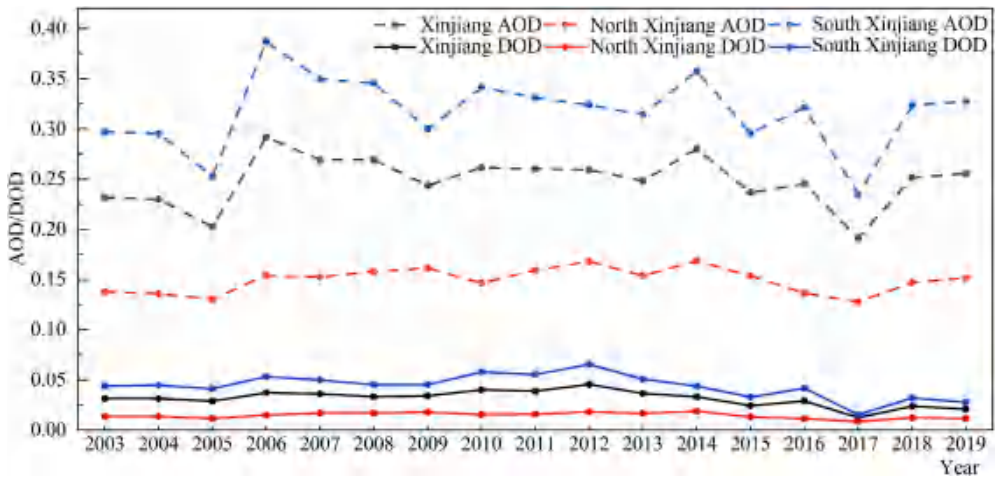

Fig. 8. Interannual variations in AOD and DOD over the Xinjiang region.

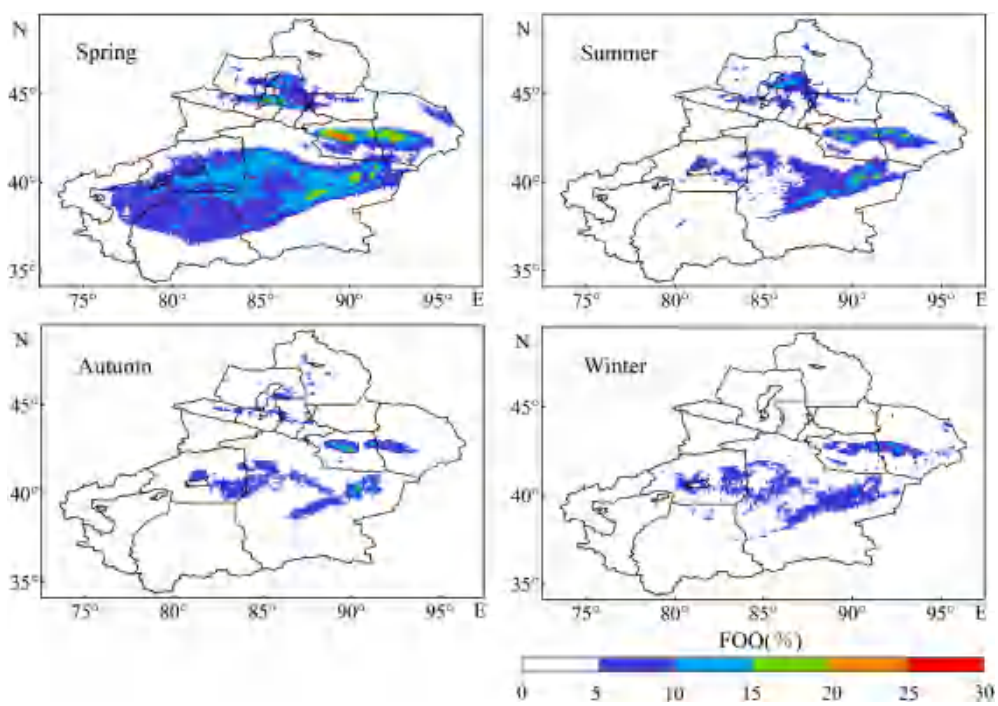

Fig. 9. Seasonal variation in the FOO distribution of DOD (DOD>0.2).

3.4. Estimation of dust emission

The dust discharged from a source area of dust is not only deposited in the source area and surrounding areas but is also transmitted over long distances and can even affect the ocean. Therefore, to better analyze the impact of dust aerosols on the environment, it is necessary to perform a quantitative study on the amount of dust emissions in the source area of dust aerosols. The seasonal average dust emissions calculated by the formula are sorted from large to small as follows: 406.36 kg m^{-2} yr^{-1}(spring) > 214.05 kg m^{-2} yr^{-1}(winter) > 153.84 kg m^{-2} yr^{-1}(summer) > 21.69 kg m^{-2} yr^{-1}(autumn). The amount of dust emissions in spring and winter is significantly greater than summer and autumn. This difference is related to the frequent dust activities in spring and polluting dust aerosols, such as urban aerosols and smoke aerosols mixed in the dust. Fig. 12 shows the seasonal average distribution of dust emissions under natural and human-made surface conditions in Xinjiang. High value centers of natural dust emissions appear in the northern Hami region and the central and eastern regions of the Tarim Basin. The average values in the four seasons are 380.91 kg m^{-2} yr^{-1}(spring), 145.85 kg m^{-2} yr^{-1}(summer), 19.02 kg m^{-2} yr^{-1}(autumn) and 205.53 kg m^{-2} yr^{-1}(winter). High values of anthropogenic dust emissions occur in the oasis on the edge of the Taklimakan Desert and in the piedmont oases of the northern slope of the Tianshan Mountains and the Turpan Basin. This result is primarily because these areas are areas with concentrated development of agriculture, industry and construction in Xinjiang, and frequent production activities cause high emissions of human-made dust. From a regional perspective, the high values of anthropogenic dust emissions in North Xinjiang appear in spring, and South Xinjiang also has high values in winter in addition to the high values in spring. These results may be related to the difference in snow cover between North and South Xinjiang. The seasonal characteristics of anthropogenic dust in the entire area are 25.45 kg m^{-2} yr^{-1} (spring) > 8.51 kg m^{-2} yr^{-1} (winter) > 7.99 kg m^{-2} yr^{-1} (summer) > 2.67 kg m^{-2} yr^{-1} (autumn). Agricultural activities produce more dust in spring, and

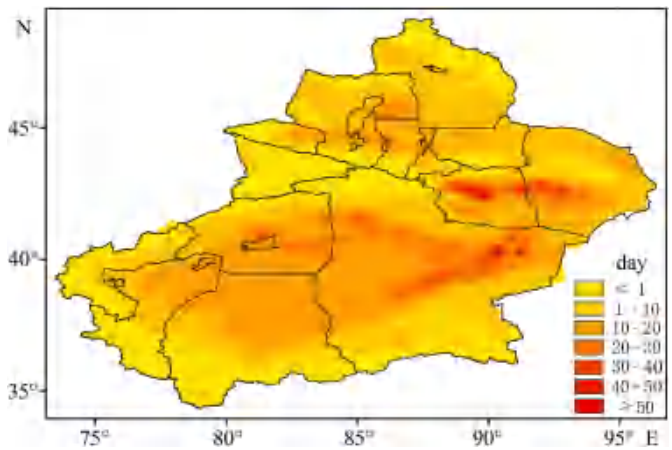

Fig. 10. Spatial distribution of dust days in Xinjiang during 2003–2019.

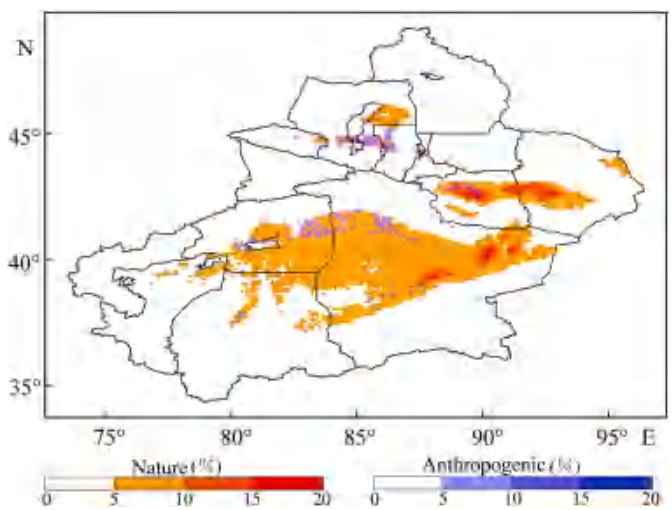

Fig. 11. Frequency distribution of anthropogenic and natural surface dust during different periods.

the anthropogenic dust in the other seasons is primarily produced through processes such as transportation and construction. Therefore, anthropogenic dust does not show obvious seasonal changes.

The distribution of average anthropogenic and natural dust emissions from 2003 to 2019 is presented in Fig. 13. The annual average of dust emissions in Xinjiang is 294.01 kg m^{-2} yr^{-1}, and natural dust aerosols exhibit a strip-like distribution, with an annual average of 271.27 kg m^{-2} yr^{-1}, which accounts for 92% of the annual dust emissions. The annual average of anthropogenic dust emissions is 22.74 kg m^{-2} yr^{-1}, which accounts for only 8% of the total dust emissions. These results indicate that natural dust aerosols dominate the dust aerosols in the Xinjiang region.

The annual average emissions of anthropogenic dust and natural dust in South and North Xinjiang and the relative contributions of anthropogenic dust and natural dust in the two regions are shown in Table 2. The natural dust emissions value in South Xinjiang is 207.96 kg m^{-2} yr^{-1}, which is three times the natural dust emissions in North Xinjiang (63.31 kg m^{-2} yr^{-1}). The average anthropogenic dust emissions in North Xinjiang (13.75 kg m^{-2} yr^{-1}) are greater than those in South Xinjiang (8.99 kg m^{-2} yr^{-1}). The highest contribution rate of anthropogenic dust in North Xinjiang is 17.84%, which indicates that human activities, such as automobiles, industrial emissions, and transportation, are more common in North Xinjiang than in South Xinjiang and led to a high level of anthropogenic dust emissions in Xinjiang. The contribution of anthropogenic dust in South Xinjiang is only 4.14%, and the dust in this region primarily comes from natural dust sources.

3.5. Factors that influence dust aerosols

3.5.1. Precipitation

The average precipitation data from 2003 to 2015 were obtained from the interpolation of data from meteorological stations across Xinjiang, as shown in Fig. 14. The annual average precipitation in the entire region varies from 29 mm to 1000 mm. The entire area was divided into arid areas (average annual precipitation of less than 200 mm), semiarid areas (200–600 mm), semihumid areas (600–800 mm) and humid areas (greater than 800 mm).

Among the different precipitation zones, the arid areas are primarily distributed in the Tarim Basin, Jungar Basin and Turpan Hami Basin and occupy the largest area, accounting for 54% of the total area. Semihumid and humid areas are mainly located in high-elevation mountainous areas, occupying the smallest area of approximately 2%. The semiarid areas are located in the middle- and low-elevation areas between the mountains and the basins and account for 44% of the total area of Xinjiang.

The trends of anthropogenic and natural dust emissions with precipitation are shown in Fig. 15. The highest emissions of natural dust aerosols (219.61 kg m^{-2} yr^{-1}) appear in extremely arid areas where the annual average precipitation is less than 100 mm yr^{-1}, and the highest anthropogenic dust emissions (9.96 kg m^{-2} yr^{-1}) appear in the area with an average annual precipitation of 100 mm–200 mm. This difference is mainly because the extremely dry areas are not suitable for human living or production activities. The natural dust emitted from arid areas accounts for 94% of the natural dust emissions in Xinjiang, and the anthropogenic dust emitted from arid areas accounts for 70% of Xinjiang's anthropogenic dust emissions, indicating that natural and anthropogenic dust primarily originates from arid areas. From arid areas to semiarid areas, the amount of dust emissions exhibits a clear decreasing trend, and the natural and anthropogenic dust emissions in the semihumid areas are near zero. This result may be related to precipitation and vegetation coverage.

3.5.2. Topography

Topography is an important factor that affects the distribution of dust. To further understand the unique topographic features of different dust sources, DEM data and Google images were used to perform targeted research on the high FOO areas of stable dust in Xinjiang. The lines of different colors in Fig. 16 correspond to the areas where the average annual FOO over 17 years is greater than 5% (dark blue), 10% (light blue), and 15% (yellow). Fig. 16 shows that the distribution of dust sources is highly correlated with the DEM data. Lower elevations exhibit higher the frequencies of dust occurrence, and this phenomenon is particularly obvious in basin areas. For example, the Aiding Lake area in the middle of the Turpan Basin has the lowest elevation in the country, and there is a natural dust source with an average annual number of dust days of 67 days. The high incidence of dust in these low-elevation areas may be due to terrain, which makes it difficult for dust to be transported outside the basin.

Using 17 years of satellite remote sensing data, the hot-spot areas of the three major dust sources in the three major basins were identified. The Lop Nur dust source area (Fig. 16e) is located on the southwestern edge of Lop Nur playa lake, where the dust frequency of occurrence is highest (55–62 days). This area is mostly covered by salt crust, and it is one of the most famous playa lakes in the world. The dust source on the southeastern edge of the Taklimakan Desert (Fig. 16d) is located at the intersection of the Algin Mountain piedmont alluvial fan and the edge of the Taklimakan Desert. The Gobi Desert in the southern Turpan Basin (Fig. 16b) has 58–62 dust days annually. The alluvial plain in the middle of the Hami Basin (Fig. 16c) is an area with a high incidence of dust, with an average of 56 days of dust per year. The areas with a high frequency

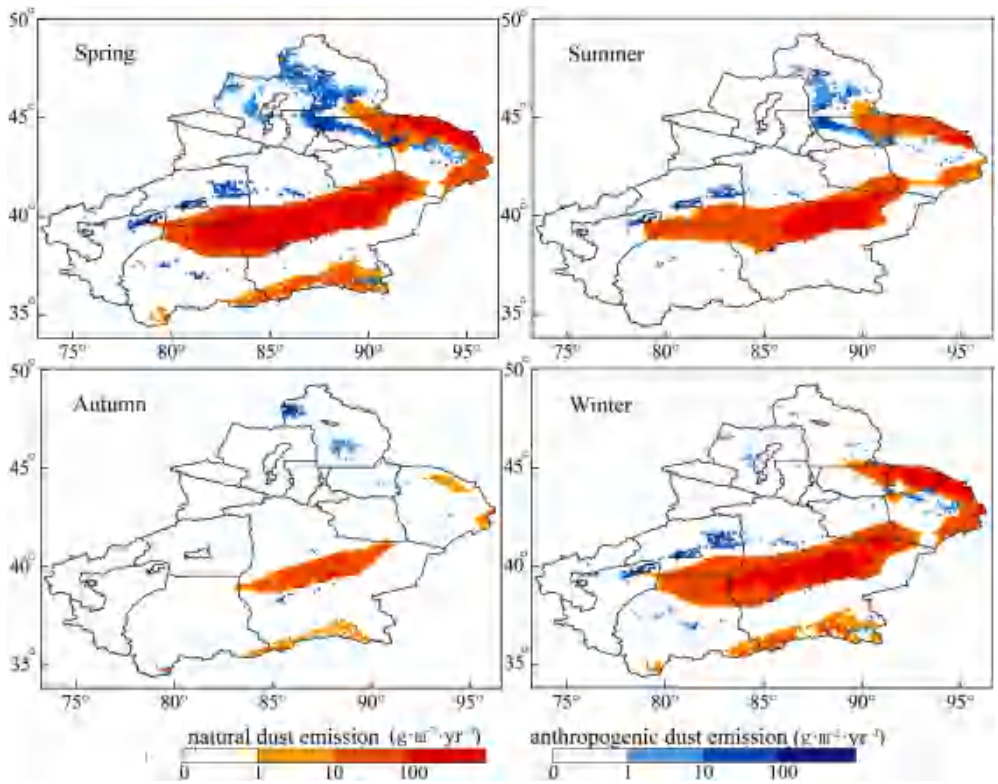

Fig. 12. Spatial distribution of average dust emissions in different seasons.

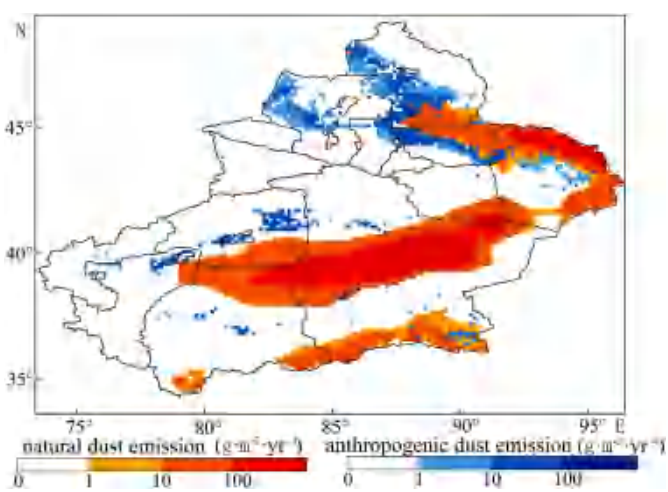

Fig. 13. Distribution of annual mean dust emissions in Xinjiang.

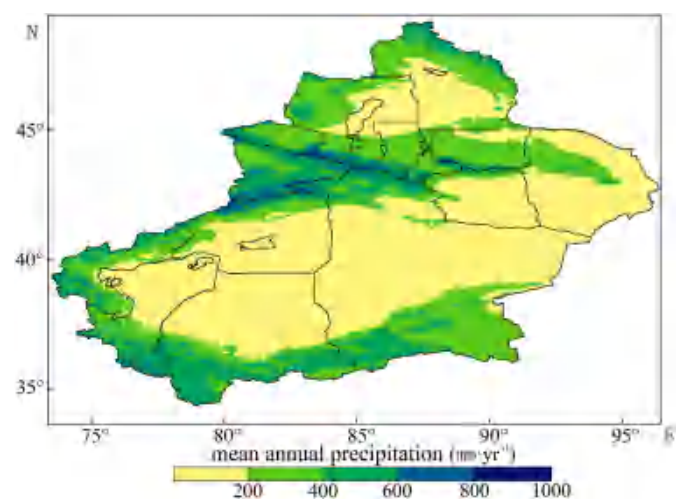

Fig. 14. Distribution of annual mean precipitation in Xinjiang.

Table 2
Natural and anthropogenic dust emissions and their contributions in Xinjiang.

	North Xinjiang	South Xinjiang
Natural dust emissions (kg m^{-2} yr^{-1})	63.31	207.96
Anthropogenic dust emissions (kg m^{-2} yr^{-1})	13.75	8.99
Total dust emissions (kg m^{-2} yr^{-1})	77.06	216.95
Contribution of anthropogenic dust (%)	17.84	4.14

of dust around the Gurbantunggut Desert (Fig. 16a) are located in the northwestern margin of the Gobi Desert and farmlands in Manas County. The average annual number of dust days is approximately 36 days.

3.5.3. *Human activities*

Although natural dust dominates the Xinjiang region, the level of urbanization is also rapidly increasing, and air pollution caused by human factors cannot be ignored. DMSP/OLS nighttime light data and GPWv4 population data were used to study the impact of human disturbance on the temporal and spatial changes in dust emissions. Nighttime light data can reflect the intensity and distribution of human activities in the area. Due to the limited time scale of the data, the 2013 nighttime light distribution (Fig. 17a) was selected for comparison with the annual average dust emission distribution (Fig. 14). The densely populated and economically developed Tianshan North Slope Economic Zone is a high-emission area of anthropogenic dust, and the digital number (DN) value of the light data is relatively large. The DN value is low in South Xinjiang, where the cities are dispersed and sparsely

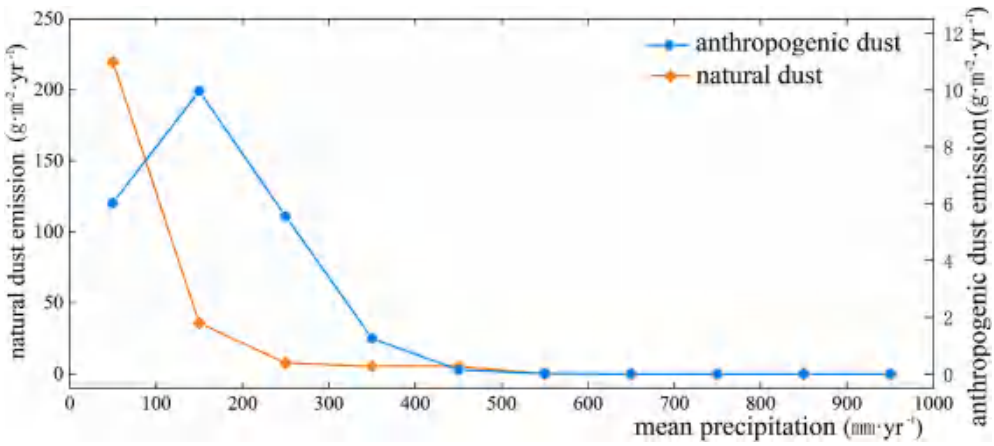

Fig. 15. Trend of anthropogenic and natural dust emissions with precipitation.

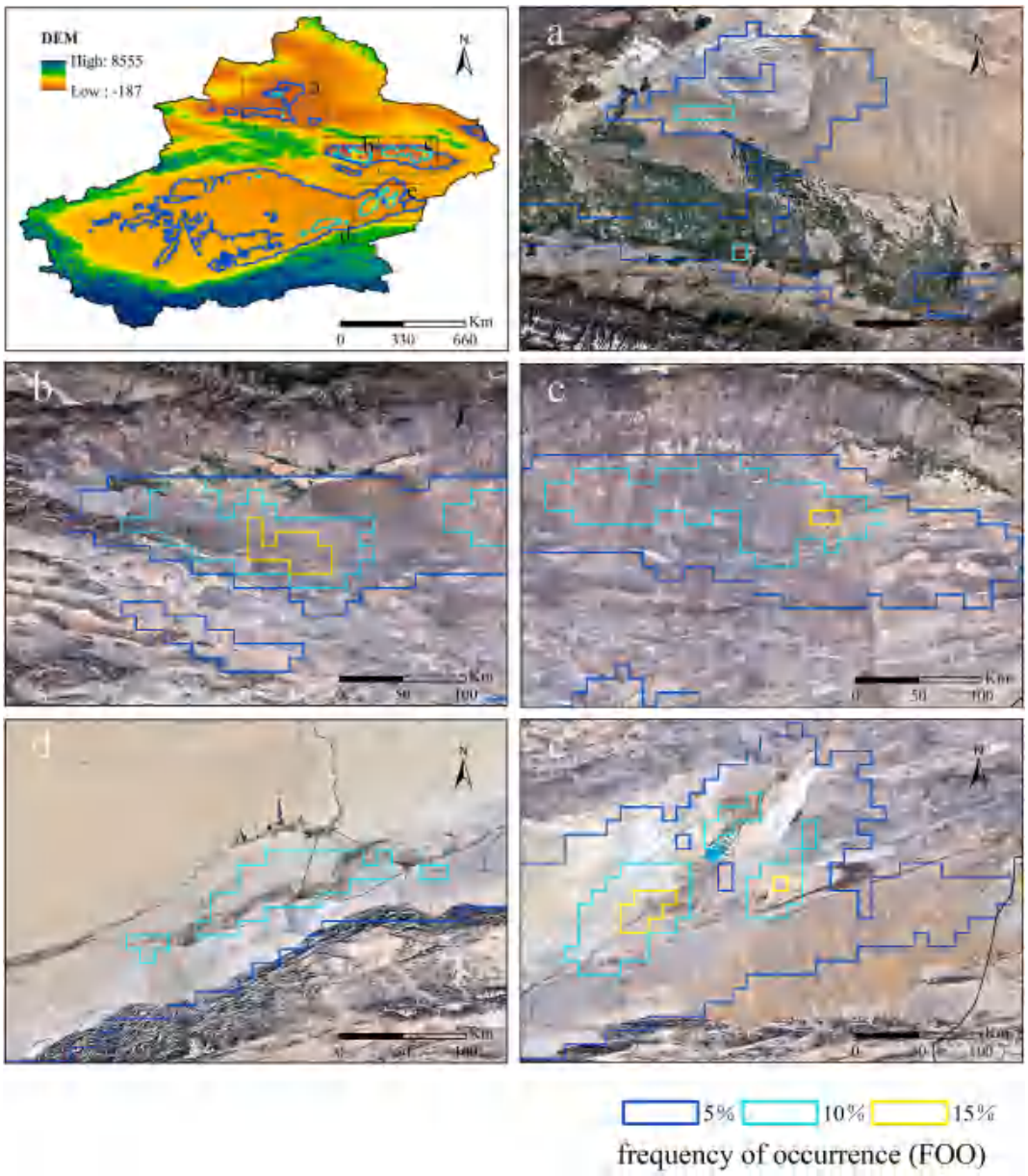

Fig. 16. Geomorphological features of the dust source region in Xinjiang.

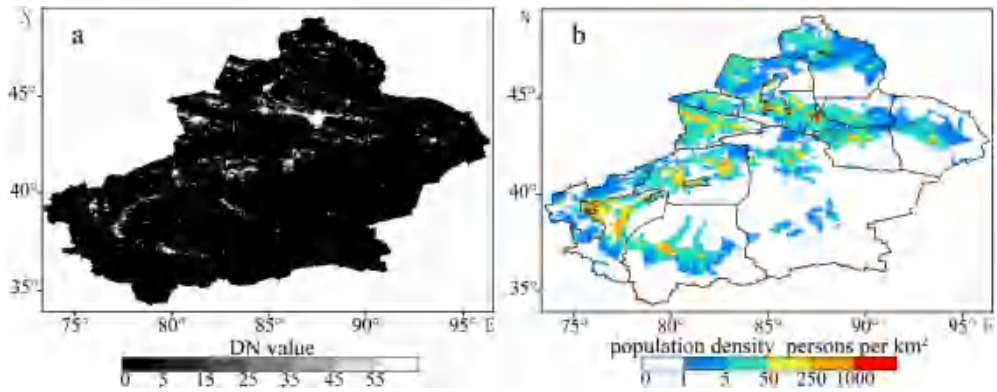

Fig. 17. Distribution of DMSP/OLS nighttime light in 2013(a) and GPWv4 population density in 2015(b).

populated, and there are fewer high-emission areas of anthropogenic dust. The nighttime light data of prefecture-level cities and county-level cities in Xinjiang from 2003 to 2013 were obtained from regional statistics and divided into two time periods with 2008 as the node. Further analysis of the correlation between the nighttime light data and the anthropogenic dust emissions in the corresponding areas and time periods showed that the two datasets are correlated, and the correlation coefficients in the two time periods of 2003–2008 and 2008–2013 are 0.648 and 0.603, respectively. The average correlation coefficient in the 11 year period is 0.502, which indicates that human social and economic activities have a positive effect on anthropogenic dust emissions.

The GPWv4 population density distribution in 2015, as shown in Fig. 17b, shows that the urban agglomeration on the northern slope of the Tianshan Mountains had the highest population density, exceeding 2000 people/km^2, followed by Kashgar and Aksu in South Xinjiang. The distribution of high population density areas is similar to the distribution of high dust emission areas in Fig. 14. Quantification of the average anthropogenic dust emissions in different regions of Xinjiang by district and matching them with the average population density of the corresponding area can reveal the relationship between population density and anthropogenic dust emissions in different regions (Fig. 18). With the slow increase in population density, anthropogenic dust emissions also show an upward trend. These results show that the change in population density has an impact on the amount of human-induced dust emissions from the surface. Larger populations increase the scale of local transportation, agriculture, animal husbandry, industry, and production activities, which are accompanied by an increase in dust emissions.

4. Conclusions

Dust aerosols can directly affect the energy balance of the global atmospheric system, and indirectly affect the microphysical characteristics of clouds and precipitation process. Xinjiang has typical sources of dust aerosols for an arid region. Based on ground-based remote sensing data and reanalysis data, the temporal and spatial distributions of dust sources in Xinjiang were studied, natural and anthropogenic dust sources were accurately identified, the amount of dust emissions was calculated, and factors that influenced dust aerosols were discussed. The following research conclusions were obtained.

(1) The MODIS DB AOD data have a high correlation with the measured AOD data at the base station site, and the extracted DOD data also exhibit good accuracy. Both types of data are suitable for dust aerosol research in Xinjiang. Daily DOD data from 2003 to 2019 were calculated and extracted, and the characteristics of temporal and spatial changes were analyzed. The results show that the highest DOD values in Xinjiang appear around the Zhungeer Basin and Tarim Basin and that the DOD values are affected by the underlying surface and climate, with a distribution characterized by significant DOD peaks in the basins. The DOD has obvious seasonal and regional differences, and high values of DOD appear in March and April. The DOD values exhibit the patterns of spring > summer > winter > autumn and South Xinjiang > North Xinjiang.

(2) The source of dust was identified by calculating the frequency of dust occurrence. There are large areas of dust sources in the three major basins of Xinjiang in spring, and the average annual frequency of dust activities is greater than 18 days. The average annual number of dust days in the Lop Nur area is highest at 84 days. The area of sand sources in the entire region shows a significant shrinking trend after spring. Natural dust sources are primarily distributed in deserts, gobi areas, dry lakes and rivers, and the range of sand sources varies little, but anthropogenic dust sources are primarily cultivated land, which changes frequently over time.

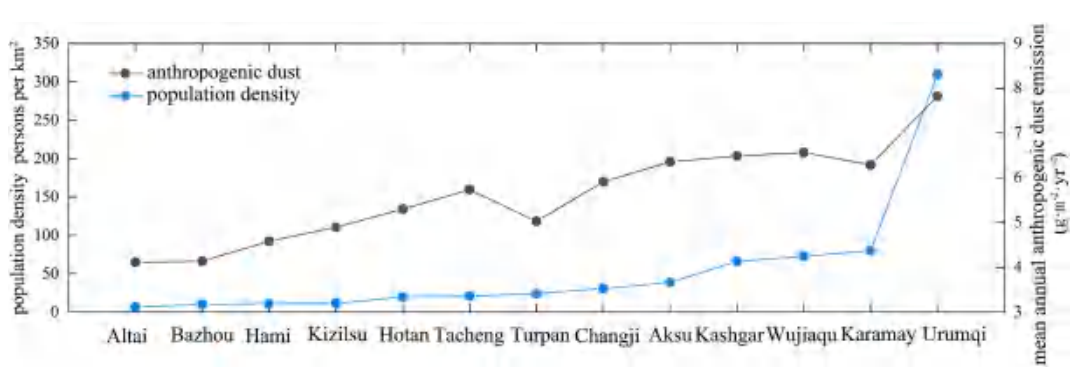

Fig. 18. Relationship between population density and anthropogenic dust emissions.

(3) Based on the calculated amount of dust emissions, the average annual dust emissions value in Xinjiang is 294.01 kg m^{-2} yr^{-1}, with natural dust aerosols accounting for 92% of the annual dust emissions, and anthropogenic dust accounting for 8% of the emissions. Further analysis of the relationships between dust emissions and precipitation, topography, and population density were performed to explore the factors that influence dust aerosols, and the results show that the amount of dust emissions was the highest in arid areas with precipitation less than 200 mm (271.47 kg m^{-2} yr^{-1}), and the amount of dust emissions decreases significantly with the increase in precipitation. The ecotone zone of the piedmont alluvial fan and marginal desert is the main landform type of the perennial sand source. There are close relationships between anthropogenic dust emissions and population density and city nighttime light data, indicating that population density and the urbanization process are two of the main factors leading to the increase in anthropogenic dust in the study area.

In conclusion, the identification of anthropogenic and natural sources and estimation of dust aerosol emissions provides a further understanding of dust aerosol sources in arid regions. Guan et al. (2016) showed that arid and semiarid regions are significant source areas, generating approximately 53% of anthropogenic dust. This study used the DOD extraction methods from Sreekanth and Kumar et al. and a threshold value of the Ångström wavelength exponent less than 0.7 as a condition for extracting DOD. Bi et al. (2017) and Zhang et al. (2019) classified particles with an Ångström wavelength exponent less than 0.2 as pure dust, and particles between 0.2 and 0.6 were defined as polluted dust that is dominated by dust and partially mixed with anthropogenic aerosols. Therefore, the study of natural and anthropogenic aerosol identification remains a field of continuous exploration and research.

CRediT authorship contribution statement

Jie Liu: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Jianli Ding:** Validation, Writing – review & editing, Supervision, Funding acquisition. **Mayila Rexiding:** Formal analysis, Investigation, Writing – review & editing, Visualization. **Xiaohang Li:** Methodology, Formal analysis. **Junyong Zhang:** Writing – review & editing. **Si Ran:** Formal analysis. **Qingling Bao:** Analysis tools. **Xiangyu Ge:** Analysis tools.

Declaration of competing interest

None.

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